Lightweight Turn-Level Hierarchical Regression with Leakage-Free Training for PHQ-8 Depression Severity Estimation

Anonymous Author(s)

Anonymous Institution

Abstract. We propose a small model that estimates depression severity from clinical interview transcripts, and a training recipe in which no model decision uses the evaluation data. Each participant turn is encoded by a frozen six-layer sentence encoder. A BiLSTM with attention reads the turn sequence and regresses the eight PHQ-8 items. The model has 1.8 M trainable parameters. Checkpoints are chosen on a holdout drawn from the training split, predictions are recalibrated with an affine map fitted on training data, each configuration must beat a shuffled-label null before it is tested, and each test run is pre-registered. On the official DAIC-WOZ test split ($n = 47$, five seeds), the model reaches Pearson $r = 0.687 \pm 0.045$, $R^2 = 0.461 \pm 0.069$, MAE 3.64 ± 0.30 and AUC 0.806 ± 0.040, The development split was not used for any decision. When checkpoints are instead chosen on the development split, as in prior work, the model scores macro-F1 0.756 and MAE 3.66, against 0.739 and 3.78 for the best published transcript-only system, with a smaller encoder and without interviewer speech. On the development split its AUC is 0.732; interview-level text, acoustic, visual and fused baselines reach at most 0.56. In ablations, using only participant turns gives a test correlation 0.140 higher than also including the interviewer, and train-fitted recalibration raises test R^2 from 0.370 to 0.461 and reduces its standard deviation across seeds by about two thirds.

Keywords: Depression severity estimation · DAIC-WOZ · PHQ-8 · Hierarchical models · Multi-task regression · Data leakage.

1 Introduction

Most work on estimating depression severity from clinical interviews is evaluated on the Distress Analysis Interview Corpus (DAIC-WOZ) [7]. The corpus pairs 189 semi-structured interviews, conducted by an animated virtual agent, with PHQ-8 self-reports [8], and ships with an official train/development/test partition [14,13].

The dataset is small. There are 107 training participants and only 30 of them are above the clinical cutoff, so results depend strongly on how the model is supervised and how the checkpoint is chosen. Evaluation practice is also uneven. A recent survey of 536 DAIC-WOZ studies [5] reports that 69.7% evaluate on

the validation split only and 81.8% report a single run. It also finds that 42.4% feed interviewer utterances to the model, which lets models use the interviewer's prompts as a shortcut [3]. When the development split both selects the model and reports its score, the estimate is optimistic by construction [15,4].

Our model builds on hierarchical symptom regression [11]. It reads only the participant's turns, encodes them with a frozen six-layer sentence encoder, and predicts the eight PHQ-8 items, so each participant provides eight graded targets instead of one binary label. In our training recipe the checkpoint, calibration and threshold are all chosen on training data, and a configuration is tested only after it beats a shuffled-label null and has been pre-registered. The contributions are:

1. A turn-level model with a frozen MiniLM encoder, a BiLSTM with attention and one regression output per PHQ-8 item. It has 1.8 M trainable parameters, and five seeds train in about 13 minutes on one T4 GPU.
2. A training recipe in which no decision uses the development or test split.
3. Test results of $r = 0.687$, $R^2 = 0.461$ and AUC 0.806. With development-set checkpointing, results match the best published transcript-only system.
4. Ablations of the input and the recipe, including held-out test results for the input choice.

2 Related Work

Interview-level models. Early challenge systems summarised each interview with functionals of acoustic (COVAREP [6]), facial and lexical features, and fed them to shallow learners [14,17]. Such summaries discard the order and local content of the conversation.

Sequence and hierarchical models. Al Hanai et al. [1] modelled interviews as sequences of question–answer segments. Mallol-Ragolta et al. [10] applied hierarchical attention networks to transcribed turns. Milintsevich et al. [11] combined frozen sentence embeddings, a BiLSTM with attention and one regression output per PHQ-8 symptom. They report test macro-F1 0.739 and MAE 3.78, the best transcript-only result we know of. Our model follows this design. It differs in its smaller encoder, its exclusion of interviewer speech, and a training procedure that does not use the development split.

Validity on DAIC-WOZ. Burdisso et al. [3] show that models given the interviewer's prompts can key on the portion of the interview where past mental-health history is discussed, which raises scores without using anything the participant says. Danylenko and Unold [5] catalogue subject leakage, missing variance and validation-only evaluation across the literature, and find that leakage-free audio models reach $R^2 = -0.058 \pm 0.063$. Our recipe avoids these problems.

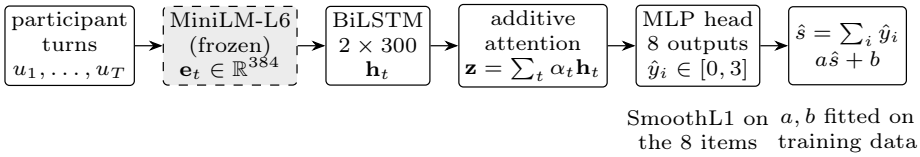


Fig. 1. Proposed model. The shaded encoder is frozen.

3 Proposed Method

3.1 Turn-Level Hierarchical Regressor

An interview is represented as the sequence of the participant's turns $(u_1, \ldots, u_T)$ after removing de-identification markers and annotation markup. Interviewer turns are excluded by design (Sect. 5.3 measures the effect). Each turn is encoded independently by the frozen `all-MiniLM-L6-v2` sentence encoder [12,16], a six-layer, 384-dimensional model, giving $\mathbf{e}_t \in \mathbb{R}^{384}$. A single-layer bidirectional LSTM with 300 units per direction reads the sequence, and additive attention [2] pools it (Fig. 1):

$$\mathbf{h}_t = \mathrm{BiLSTM}(\mathbf{e}_{1:T})_t, \qquad \alpha_t = \mathrm{softmax}_t\big(\mathbf{v}^\top \tanh(\mathbf{W}\mathbf{h}_t)\big), \qquad \mathbf{z} = \textstyle\sum_t \alpha_t \mathbf{h}_t. \tag{1}$$

A head (linear, LeakyReLU, LayerNorm, dropout, linear) maps $\mathbf{z}$ to eight outputs, each clamped to the item range $[0, 3]$. The severity estimate is their sum $\hat{s} = \sum_{i=1}^{8} \hat{y}_i$, and the binary screening decision is $\mathbb{1}[\hat{s} \geq 10]$.

Item-level multi-task supervision. The model is trained only with a SmoothL1 loss on the eight item scores; there is no classification loss. We chose this because labels are scarce. A binary objective has 30 positive examples in the training set and a total-score objective has one value per participant. Item-level regression has $8 \times 106 = 848$ graded targets from the same interviews, and the eight outputs share one interview representation.

Frozen, compact encoder. Freezing the encoder keeps the number of trainable parameters small (1.80 M) for 106 training participants. It also means the encoder cannot carry label information between splits: MiniLM was pretrained on general text, never sees a PHQ-8 label, and is applied the same way to every split.

3.2 Leakage-Free Training Recipe

Train-holdout checkpointing. In each seed, 20% of training participants are held out and the checkpoint with the highest holdout correlation is kept. The development split is not used, so development scores can be used to compare models without bias.

Train-fitted recalibration. Regression towards the mean compresses predicted totals. We fit $s \mapsto a\hat{s} + b$ by least squares on the training predictions and apply it unchanged to the development and test splits. Correlation and AUC are invariant to this map; R^2, MAE and threshold metrics are not.

Null gating. A configuration is evaluated on test only if it beats a ten-seed null on development data. In the null, training targets are shuffled across participants and training, checkpointing and recalibration are repeated unchanged.

Pre-registered test evaluation. Each test evaluation is declared in a time-stamped record (configuration, seeds, hyper-parameters) committed before the run, and the code refuses to re-evaluate a configuration whose test result exists.

Training details. AdamW [9] with learning rate 10^{-3}, weight decay 10^{-2}, dropout 0.3, gradient clipping at 1.0, 200 full-batch epochs, five seeds. The learning rate is higher than for fine-tuned encoders because gradients do not flow into the frozen encoder.

4 Experimental Setup

Data. We use all 189 DAIC-WOZ participants with the official split [13], read from the released split files: train 107 (30 depressed), development 35 (12) and test 47 (14). Labels are the released PHQ-8 scores and binary labels, used verbatim. Item scores exist for train and development only, so test predictions are scored on the total and the binary label. One training participant lacks item labels, leaving 106. The cleaned transcripts contain 31,525 participant turns (mean 167 per interview, range 42–385).

Baselines. We build interview-level baselines on the same participants. *Text*: TF-IDF over the participant transcript (400 uni- and bigrams) reduced by truncated SVD to 40 dimensions. *Audio*: functionals of COVAREP and formant features, with source features aggregated over voiced frames and frames masked to participant speech. *Visual*: functionals of facial action units, gaze and head pose. *Fusion*: their concatenation. Each is evaluated with L1/L2 logistic regression and random forests, with feature selection and hyper-parameters tuned by nested cross-validation inside the training split [15]. We also compare with the published results of Milintsevich et al. [11].

Protocols. Our primary protocol is the recipe of Sect. 3.2. For comparability with published work we also report the *standard* protocol, in which the checkpoint is chosen by development-set F1 and results are reported on test, as in [11].

Metrics. We report Pearson r, R^2, MAE and AUC on the PHQ-8 total, and macro-F1 at the clinical cutoff. All are given as mean $\pm$ standard deviation over five seeds. We do not use positive-class F1 as a primary metric: on the development split, labelling every participant as depressed already scores 0.511, above both official AVEC 2016 baselines [14].

Table 1. Comparison with interview-level baselines ($n = 35$). †Against the shuffled-label null.

model	dev AUC	train CV AUC	perm. p
text: TF-IDF + SVD	0.464	0.471	0.652
audio: COVAREP/formant funct.	0.558	0.631	0.090
visual: AU/gaze/pose funct.	0.482	0.382	0.920
early fusion	0.471	0.625	0.104
proposed (turn-level)	$\mathbf{0.732 \pm 0.078}$	—	$0.0063^{\dagger}$

Table 2. Test results ($n = 47$, five seeds). cal.: recalibrated.

	proposed recipe (checkpoint: train holdout)		standard protocol (checkpoint: development)	
	raw	cal.	raw	cal.
r	$\mathbf{0.687 \pm 0.045}$		0.727 ± 0.059	
AUC	$\mathbf{0.806 \pm 0.040}$		0.834 ± 0.049	
R^2	0.370 ± 0.199	$\mathbf{0.461 \pm 0.069}$	0.491 ± 0.122	0.525 ± 0.089
MAE	3.74 ± 0.47	$\mathbf{3.64 \pm 0.30}$	3.66 ± 0.59	3.48 ± 0.34
macro-F1	0.637 ± 0.107	$\mathbf{0.679 \pm 0.041}$	0.756 ± 0.066	0.765 ± 0.061

5 Results

5.1 Comparison with Interview-Level Baselines

Table 1 compares the model with the interview-level baselines on the development split. All models were trained on the training split only, and none was selected on development data. The best interview-level model reaches development AUC 0.558 (audio). The proposed model reaches 0.732 ± 0.078 using the same transcripts as the text baseline, which itself scores below chance. Permutation tests on the training split (200 shuffles, fixed a-priori pipeline) find no detectable signal in any interview-level representation ($p \geq 0.090$). Our model beats its shuffled-label null on correlation ($p = 0.0007$), recalibrated R^2 ($p = 0.0003$) and AUC ($p = 0.0063$). All three remain significant after Bonferroni correction.

5.2 Held-Out Test Performance

Table 2 reports the pre-registered test results. With the proposed recipe, the model reaches $r = 0.687 \pm 0.045$ (Fisher 95% CI at $n = 47$: $[0.498, 0.814]$; $p \approx 10^{-7}$), AUC 0.806 ± 0.040, recalibrated $R^2 = 0.461 \pm 0.069$ and an MAE of 3.64 PHQ-8 points. At the fixed cutoff, mean sensitivity is 0.50 and specificity 0.85. This pattern is consistent with a pure regression objective: the model ranks severity well but was never trained to place the cutoff.

Table 3. Comparison with Milintsevich et al. on the test split.

	Milintsevich et al. [11]	proposed (standard protocol)
test macro-F1	0.739	0.756 ± 0.066
test MAE	3.78	3.66 ± 0.59
sentence encoder	RoBERTa-based (S-RoBERTa)	MiniLM-L6, 384-d, frozen
input turns	not documented	participant only
variance reported	no (5-seed mean)	yes (5 seeds, mean ± SD)

Table 4. Ablations (five seeds; bold: proposed; —: not tested).

		development ($n = 35$)		test ($n = 47$)	
component	variant	r	R^2	r	R^2
input turns	participant only	0.533 ± 0.078	0.204 ± 0.087	$\mathbf{0.687 \pm 0.045}$	$\mathbf{0.461 \pm 0.069}$
	+ interviewer	0.577 ± 0.137	0.221 ± 0.128	0.547 ± 0.132	0.293 ± 0.125
recalibration	none		0.074 ± 0.271		0.370 ± 0.199
	holdout-fitted	(invariant)	0.111 ± 0.076	(invariant)	—
	train-fitted		$\mathbf{0.204 \pm 0.087}$		$\mathbf{0.461 \pm 0.069}$
		macro-F1: observed / null			
threshold	fixed (PHQ-8 $\geq$ 10)	0.514 / 0.424		—	
	holdout-selected	0.583 / 0.495		—	

Comparison with the state of the art. Under the standard protocol, which matches the evaluation of Milintsevich et al. [11], our uncalibrated results are macro-F1 0.756 ± 0.066 and MAE 3.66 ± 0.59 against their reported 0.739 and 3.78 (Table 3). The differences are a quarter and a fifth of one seed standard deviation, so the two systems perform about the same. Ours uses a six-layer, 384-dimensional encoder instead of a RoBERTa-based one, and leaves out the interviewer's turns, about a third of the dialogue.

5.3 Ablation Study

Table 4 varies the input and each part of the training recipe, one at a time. Test results exist only for pre-registered configurations.

Participant-only input. On development data, adding interviewer turns changes r by $+0.044$, which is not significant. On the test split the effect is negative. Including the interviewer lowers test r by 0.140 ± 0.114 ($p = 0.052$, paired over seeds) and recalibrated R^2 by 0.168 ± 0.118 ($p = 0.034$), and roughly triples the seed standard deviation of r. This fits the finding that interviewer prompts give models a shortcut that does not carry over to new participants [3].

Recalibration. Train-fitted recalibration raises test R^2 from 0.370 to 0.461 and cuts its seed standard deviation from 0.199 to 0.069. It also lowers test MAE from 3.74 to 3.64. Fitting the same map on the train-internal holdout ($n \approx 21$) is

worse than fitting it on the full training split, because the fitted slope is unstable at that size. It averages 0.56 under real labels and 9.39 under shuffled labels.

Threshold. Selecting the decision threshold on the holdout raises development macro-F1 by 0.069, but raises the shuffled-label null by 0.071. The gain comes from the selection procedure, so we keep the clinical cutoff fixed.

Checkpoint criterion. Selecting checkpoints on the development split instead of the train holdout (the standard protocol) improves test r by 0.040 ± 0.093 and macro-F1 by 0.119 ± 0.140: the 35 extra labelled participants help choose the checkpoint. Development scores then become optimistic, however. Development-set checkpointing raises development R^2 by 0.233 but test R^2 by only 0.121, so about half of the apparent gain does not generalise. We choose checkpoints on the training holdout so that the development split can still be used for comparisons such as Table 1.

5.4 Sanity Check: Reproducing a Published Negative Result

To check that the pipeline does not leak, we fitted ridge regression on the interview-level audio functionals. It yields development $R^2 = -0.060$ ($r = -0.067$, $p = 0.70$), within 0.002 of the -0.058 ± 0.063 reported for leakage-free audio models by Danylenko and Unold [5] with a different architecture. The pipeline therefore gives the expected null result for audio functionals.

6 Discussion and Limitations

The same transcripts that yield no signal as an interview-level bag of words yield test $r = 0.687$ when modelled as a sequence of turns. Two changes could explain this: keeping the turn structure, and using a pretrained encoder instead of TF-IDF. Our experiments do not separate them; an interview-level model over averaged MiniLM embeddings would. The item-level heads, attention and BiLSTM were chosen from prior work and have not been ablated. The released code includes options for these ablations.

DAIC-WOZ is small. With 106 training participants every estimate has a wide interval, and five seeds give the held-out ablations little power. We use a single corpus, interview protocol and language, with no external validation. The test split was opened three times, once per pre-registered configuration (proposed recipe, standard protocol, interviewer ablation), and only the first is our primary result. We decided to recalibrate after seeing unstable development R^2. The fit uses training data only and the null was rerun with the same procedure, but the decision was post hoc. Classification at a fixed cutoff is not reliably estimable at development-set size. PHQ-8 is a self-report screening instrument, and nothing here supports clinical use.

7 Conclusion

We presented a turn-level hierarchical regressor for PHQ-8 severity estimation, trained so that no decision uses the evaluation data. With a frozen six-layer encoder and 1.8 M trainable parameters, it reaches test $r = 0.687$ and AUC 0.806. With development-set checkpointing it matches the best published transcript-only system, and its development AUC is well above that of interview-level baselines. Using only participant turns and recalibrating on training data both improved test accuracy. Code, per-seed results and pre-registration records are released with the paper.

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